SECTION 13851-A
FIRE ALARM AND SMOKE DETECTION SYSTEMS

PART 1 – GENERAL

1.1 WORK INCLUDED

A. Furnish and install a complete and operational fire alarm and smoke detection system.

1.2 RELATED WORK

A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for fire alarm and smoke detection systems:
   1. Section 13855-A – Very Early Warning Smoke Detection System.
   2. Section 13930-A – Automatic Sprinkler Systems.
   3. Section 16010-A – General Electrical Provisions.

B. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements.

C. In the event of conflict regarding fire alarm and smoke detection system requirements between this Section and any other section, the provisions of this Section shall govern.

1.3 SYSTEM DESCRIPTION

A. The fire alarm system shall be a proprietary system as defined in NFPA 72.

B. Contractor to ensure that the software and hardware to be installed shall be capable with data functionality as year-2000 compliant.

C. System Supervision: Provide electrically supervised system with supervised alarm-initiating and alarm-indicating circuits. Network communication circuits shall be Style 7 type and device communication loop wiring shall be Style D type. Occurrence of single-ground or open condition in initiating or signaling circuit places circuit in trouble mode. Component or power supply failure places system in trouble mode.

D. Alarm Reset:
   1. Manual acknowledge function at control panel silences audible alarm/trouble alarm; visual alarm at panel is displayed until initiating alarm/trouble is cleared.
   2. Reset function resets alarm system out of alarm if alarm-initiating circuits have cleared.

E. Trouble Sequence of Operation - System trouble, including grounding or open circuit of supervised circuits or power or system failure, causes system to enter trouble mode, including the following operations:
   1. Visual and audible trouble alarm by device on fire alarm control panel and on remote graphic display.
   2. Transmit trouble signal to Security Command Center.
   3. Manual acknowledge function at Security Command Center silences audible trouble alarm; visual alarm is displayed until initiating trouble is cleared.

F. Lamp Test: Manual lamp test function causes alarm and trouble indication at fire alarm control panel.

G. The fire alarm system shall be interfaced with the Very Early Warning Smoke Detection (VEWSD) system as specified in Section 13855-A, Very Early Warning Smoke Detection System.

1.4 OPERATION AND MAINTENANCE DATA

A. Include manufacturer's letter stating that system is operational.

B. Provide as-installed schematic, wiring, plan view equipment layout, and point-to-point interconnection diagrams of all circuits, internal and external, for all equipment installed. Schematics shall include the conductor color-coding system.

C. Provide fire alarm control panel internal wiring diagram with function description for incoming and outgoing terminal.

PART 2 -- PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Owner approval.

B. Fire alarm system contractor and cleanroom contractor shall have the same manufactures/product to integrate both portions as a completed fire alarm system..

2.2 FIRE ALARM AND SMOKE DETECTION CONTROL PANEL

A. Control Panel: control panel shall be intelligent type, modular construction with surface wall-mounted enclosure.

B. Power Supply: adequate to serve control panel modules, remote detectors, remote annunciators, door holders, smoke damper controllers, relays, and alarm-indicating devices. Include battery-operated emergency power supply with capacity for operating system in standby mode for 4 hours followed by alarm mode for 10 minutes. Batteries shall be sealed, maintenancefree type.

2.3 INITIATING DEVICES

A. Manual Pull Station:
   1. Taiwan local code standard.

B. Heat Detector:
   1. Fixed temperature.
   2. Combination rate of rise and fixed temperature.

C. Elevator Equipment Room Heat Detector:
   1. Fixed temperature 135 degrees F.

D. Ceiling-Mounted Smoke Detector:
   1. Photoelectric type.
   2. Ionization type.

E. Duct-Mounted Smoke Detector:
   1. Photoelectric type with duct sampling tubes extending the width of duct and visual indication of detector actuation in duct-mounted housing.

F. Flame Detector:
   1. Ultraviolet/infrared radiation type, suitable for Class 1, Division 2 installation.

2.4 INDICATING DEVICES

A. Strobe light: Taiwan local code standard.

B. Horn/Speaker: Taiwan local code standard.

2.5 AUXILIARY DEVICES

A. Door Release:
   1. Magnetic door holder with integral diodes to reduce buzzing, 24-Vdc coil voltage.

2.6 FIRE ALARM WIRE AND CABLE

A. Fire alarm cable types shall be indicated within the drawing set and shall be in accordance with the requirement of Section 16010-A, General Electrical Provisions.

PART 3 -- EXECUTION

3.1 INSTALLATION

A. Install system in accordance with manufacturer's instructions and as shown on the drawings.

B. Mount end-of-line device in box with last device or separate box adjacent to last device in circuit.

C. Mount outlet box for electric door holder to withstand 80 pounds pulling force.

D. Make conduit and wiring connections to door release devices, sprinkler flow switches, sprinkler valve tamper switches, post indicator valves (PIV), fire suppression system control panels, damper controllers, duct smoke detectors, and selected emergency shower stations. See P&ID drawings for details.

3.2 FIELD QUALITY CONTROL

A. Test in accordance with NFPA 72 and local fire department requirements in the presence of the Owner and under the supervision of the manufacturer.

B. Acceptance of the fire alarm system by the Owner will not take place until Section 3.2.A have been completed in their entirety.

3.3 MANUFACTURER'S FIELD SERVICES

A. Include services of certified technician to supervise installation, adjustments, final connections, and system testing.

B. Provide the Owner instruction in the operation of the system, test equipment, and necessary maintenance of the detection devices.

- END -